Ruhan Bhavsar

365-998-6224 | r5bhavsa@uwaterloo.ca | LinkedIn | GitHub(Ruhan456) | ruhanbhavsar.com

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechatronics Engineering — *Expected 2028*

Experience

Embedded Systems Validation Developer (Motor Control) — Sept 2026 - Dec 2026
onsemi — *Waterloo, ON*

Engineering Automation Intern — Jan 2026 - Apr 2026
MPI Morheat — *Toronto, ON*

- Developed a **React-based** configurator for flange, screwplug, and PTFE heater drawings, reducing custom drawing generation time by **95%** from approximately **20 minutes to 1 minute** for customer proposals.
- Developed a low-cost (<**$500**) automated **three-axis label cutter** using **stepper motors**, **3D-printed components**, and an **STM32 microcontroller** programmed in **C++**, **saving 5–10 hours of manual labor per week**.

Engineering Intern — May 2025 - Aug 2025
Vexos — *Markham, ON*

- Built manufacturing tools with **Python**, **SolidWorks**, and **3D printing** that reduced equipment downtime by **40 hours monthly**.
- Automated calibration workflow in **Asset Essentials**, reducing overdue work orders by **30%** and ensuring **100% compliance** with industry standards.

Firmware Developer — Sept 2024 - Present
Midnight Sun Solar Car Team — *Waterloo, ON*

- Maintained embedded software for **8 microcontrollers**, managing critical **real-time tasks** and **ADC sampling** with a **10 ms latency target** to ensure system-wide synchronization.
- Implemented a **Hardware-in-the-Loop (HIL)** testing framework using **Pytest** and **Python scripts**, reducing manual testing time by **80**%.

Projects

Custom STM32 RTOS | *STM32F401, ARM Cortex-M4, C, ARM Thumb Assembly*

- Developed a custom **RTOS kernel** with APIs for kernel initialization, thread creation, kernel startup, and voluntary thread yielding.
- Implemented cooperative and preemptive **round-robin scheduling** using **SVC**, **PendSV**, and 1 ms **SysTick** time slices, with ARM assembly for **MSP/PSP switching** and register context restoration.
- Allocated 8-byte-aligned thread stacks from a fixed memory pool, constructed initial ARM exception frames, and passed thread arguments through **R0** according to the **ARM EABI**.

Natural Disaster Severity Mapping | *React, Flask, MongoDB, PyTorch, ResNet34*

- Built a full-stack web application using **React**, **Flask**, and **MongoDB** to assess earthquake severity from user-uploaded images.
- Trained a **ResNet34 CNN** in **PyTorch** to classify disaster severity with **95% accuracy** across **8,500 images**.
- Integrated the **Mapbox API**, secure authentication, and real-time map updates into a responsive user interface.

Arduino Emergency Responder | *Arduino, C++, Python, PySerial, Flask, React*

- Developed a full-stack emergency-response application with a **React** frontend and **Flask** backend, using **PySerial** to process real-time alerts from an Arduino.
- Designed a severity-based prioritization algorithm and interactive dashboard to rank incidents and display live emergency updates.

Technical Skills

Programming Languages: C, C++, Python, ARM Thumb Assembly, VHDL, MATLAB, JavaScript, TypeScript
Embedded Platforms: STM32F401, ARM Cortex-M4, ESP32, Arduino, RTOS, ADC
Tools/Libraries: STM32CubeIDE, Git, Linux, SCons, Vagrant, Pytest, PySerial
Concepts: Bare-Metal Development, Real-Time Systems, Context Switching, Interrupt Handling, HIL Testing, Memory Management, Unit Testing, CI/CD